

Data Quality Report (DQR)

Dataset: ppr-group-25208508-train.csv

Module: COMP47350 Group Project - Part (1) DQR

Scope: Data understanding, quality diagnosis, and action recommendations for downstream cleaning and modeling.

1. Executive Summary

This report presents a structured data quality assessment for the training dataset used in the residential property price prediction task. The objective is to identify quality risks that can affect statistical analysis and model robustness, then define practical actions for the next stage (Data Quality Plan, DQP).

The dataset contains **54,000 rows and 9 columns**. At a high level, the data is usable but requires targeted handling in four critical areas:

- **High missingness** in location/size-related fields (Eircode, Property Size Description).
- **High-cardinality text features** (Address, Eircode) that are not directly model-friendly.
- **Price outliers and heavy right skew**, which may destabilize model training.
- **4. Small number of exact duplicates** that should be removed for consistency.

Core recommendation: preserve business signal while simplifying fragile raw fields through controlled feature engineering, robust target handling, and transparent step logging in DQP.

2. Dataset Overview

2.1 Shape and Features

- Rows: **54,000**
- Columns: **9**
- Features:
 - Date of Sale (dd/mm/yyyy)
 - Address
 - County
 - Eircode
 - Price (€)
 - Not Full Market Price
 - VAT Exclusive
 - Description of Property
 - Property Size Description

2.2 Time Coverage

Date of Sale is fully populated and valid for all records. Transactions span **2016 to 2024**, with near-even annual counts in this grouped sample.

3. Data Quality Findings

3.1 Missingness

The missingness profile is concentrated in two columns.

Feature	Missing Count	Missing Rate
Property Size Description	51,164	94.75%
Eircode	37,108	68.72%
Other columns	0	0%

Interpretation: - Property Size Description has very limited usable coverage and cannot reliably support size-based modeling decisions in its current form. - Eircode is heavily missing and inconsistent as a direct location key.

3.2 Duplicates and Constants

- Exact duplicate rows: **9** (0.017% of all rows)
- No meaningful fully constant analytical column after preprocessing checks.

Interpretation: - Duplicate count is small but should still be removed to avoid duplicated training signal and metric distortion.

3.3 Cardinality and Feature Usability

High-cardinality fields:

- - Address: **53,261 unique** values
- - Eircode: **16,823 unique** values

Moderate/low-cardinality fields:

- - County: 26
- - Description of Property: 3
- - Not Full Market Price: 2
- - VAT Exclusive: 2

Interpretation: - Raw Address and Eircode should not be directly one-hot encoded. - Structured, low-dimensional derived location features are preferred.

3.4 Price Quality and Distribution

Parsed numeric Price (€) summary:

- Mean: **340,639** - Median: **250,000** - Std: **1,662,132** - Min: **5,250**
 - 5th percentile: **60,000** - 95th percentile: **703,000** - Max: **225,000,000**

Interpretation: - Distribution is strongly right-skewed. - Extreme values (especially very high-end records) can dominate model loss and coefficient estimation. - Robust transformation/clamping should be applied in DQP.

3.5 Category Balance (Key Fields)

County

- Dublin: **16,592** (~30.7%)
- Cork: 5,975
- Kildare: 2,926
- Others with smaller shares

Description of Property

- Second-Hand Dwelling house /Apartment: **44,332**
- New Dwelling house /Apartment: **9,665**
- Minor residual category: 3

VAT and Full-Market Indicators

- VAT Exclusive: Yes = **9,514**, No = 44,486
- Not Full Market Price: Yes = **2,850**, No = 51,150

Interpretation: - Category imbalance is manageable but must be acknowledged in interpretation and model evaluation. - VAT Exclusive carries business-rule implications for target standardization.

4. Visual-Analytical Interpretation (from notebook figures)

The notebook visuals support four consistent conclusions:

- **Price distribution is heavy-tailed**, and raw boxplots show long upper whiskers and extreme outliers.
- **County-level heterogeneity is significant**, especially for Dublin/Wicklow compared with many other counties. 3. **Time signal exists** (year-level trend stronger than month-level variation). 4. **High-cardinality location raw fields** are not directly suitable for robust modeling and should be transformed.

These observations justify moving from raw text location fields to engineered location features and from raw price to standardized robust target variants in DQP.

5. Recommended Actions for DQP

5.1 Feature-Level Plan

Feature	DQR Issue	Recommended DQP Action
Price (€)	Heavy skew + extreme outliers	Parse to numeric, apply business-rule standardization and robust clamping
VAT Exclusive + Description of Property	Price definition inconsistency risk	Rule-based VAT adjustment for qualifying new dwellings
Address	Very high cardinality, noisy free text	Extract compact interpretable features, then drop raw address
Eircode	High missingness + high cardinality	Drop or convert only if stable transformed signal exists
Property Size Description	94.75% missing	Drop in final cleaned training set
Date of Sale	Raw date not model-ready	Derive Sale Year and Sale Month
Duplicates	Exact duplicates present	Remove duplicate rows

5.2 Modeling-Safety Principles

- Keep preprocessing deterministic and auditable.
- Avoid leakage from target-derived transformations in predictors.

- Prefer robust target formulation for stable evaluation.
- Ensure no missing values remain in the final DQP output dataset.

6. Risk Assessment

If DQP actions are not applied, expected risks include: 1. Unstable model coefficients due to target outliers. 2. Overfitting and sparse matrices from raw high-cardinality text encoding. 3. Reduced reproducibility from ad-hoc preprocessing. 4. Weaker generalization due to unresolved price-definition inconsistency.

7. Conclusion

The training dataset is adequate for predictive modeling after targeted cleaning and transformation. The DQR confirms that the highest-impact interventions should focus on: - robust price handling, - location feature engineering, - explicit missingness strategy, - duplicate removal, - and feature-type finalization.

These decisions form the technical bridge from descriptive quality assessment (DQR) to executable cleaning policy (DQP), and they are expected to materially improve model stability and interpretability in later project stages.

8. Detailed DQR Procedure Alignment with `project.ipynb`

This section adds the detailed DQR procedural items present in `organize/project.ipynb` and requested for alignment.

8.1 Data Type Conversion (Raw -> Converted)

Feature	Raw Type	Converted Type	Notes
Date of Sale (dd/mm/yyyy)	string/object	datetime64	Parsed with <code>dayfirst=True</code> ; invalid values coerced to NaT
Price (€)	string/object	float64	Currency symbol/comma removed, then numeric parse
County	string/object	category	Low-cardinality categorical feature
Not Full Market Price	string/object	category	Binary categorical feature
VAT Exclusive	string/object	category	Binary categorical feature
Description of Property	string/object	category	Low-cardinality categorical feature
Property Size Description	string/object	category	Sparse but standardized before final drop decision
Address	string/object	string (kept raw in DQR stage)	High-cardinality field; transformed in DQP
Eircode	string/object	string (kept raw in DQR stage)	High-missing/high-cardinality field
Sale Year	N/A	Int64 (derived)	Derived from parsed sale date
Sale Month	N/A	Int64 (derived)	Derived from parsed sale date

8.2 Save Interim Cleaned Dataset (Intermediate Artifact)

After core DQR preprocessing (type parsing and duplicate handling), an interim cleaned version is saved for traceability and downstream checks before DQP rule execution.

- Purpose: reproducibility and stage-to-stage audit trail.
- Scope: still a DQR-stage artifact (not final DQP output).

8.3 Table Structure (Standalone)

To match the notebook structure, descriptive statistics are presented in three separate blocks:

- Numeric Features Summary Table
- count, mean, std, min, quartiles, max for numeric variables.
- Categorical Features Summary Table
- non-null count, unique count, top category, top frequency.
- High-Cardinality String Features
- explicit cardinality report for raw string features (notably Address, Eircode) and model-usability commentary.

8.4 VAT Consistency Check (Process Detail)

A rule-consistency check is performed at DQR stage before DQP adjustment:

- Build cross-tab between VAT Exclusive and Description of Property.
- Verify that VAT-related positive cases are concentrated in New Dwelling house /Apartment.
- Flag potential inconsistencies for DQP handling.

This provides the direct rationale for the DQP rule: - for qualifying new dwellings with VAT Exclusive = Yes, create VAT-adjusted price representation.

8.5 Export DQR Summary Tables (Action + Filenames)

The DQR summary exports are explicitly produced as CSV files.

From project.ipynb: - DQR-NumericFeatures.csv - DQR-CategoricalFeatures.csv

From standalone DQR.ipynb workflow (naming variant):

- train-cleaned.csv
- DQR-NumericFeatures.csv
- DQR-CategoricalFeatures.csv

8.6 Structured Findings (Subsection Form)

To mirror notebook structure, DQR findings are grouped as:

- **Missing Value Issues:** concentrated high missing (Property Size Description, Eircode).
- **Structural Issues:** duplicates and field usability constraints.
- **Distribution & Outlier Issues:** heavy right tail in price and extreme values.
- **Logical Consistency Issues:** VAT-related business-rule consistency.
- **Feature Usability Summary:** keep/transform/drop guidance for DQP.

Appendix A: DQR Code-Generated Figures and Tables (from `organize/project.ipynb`)

This appendix lists outputs generated by DQR code cells, with section mapping and brief descriptions.

A.1 Figures

Figure A1: Output from cell 28 (1.7 Missing Value Visualisation)

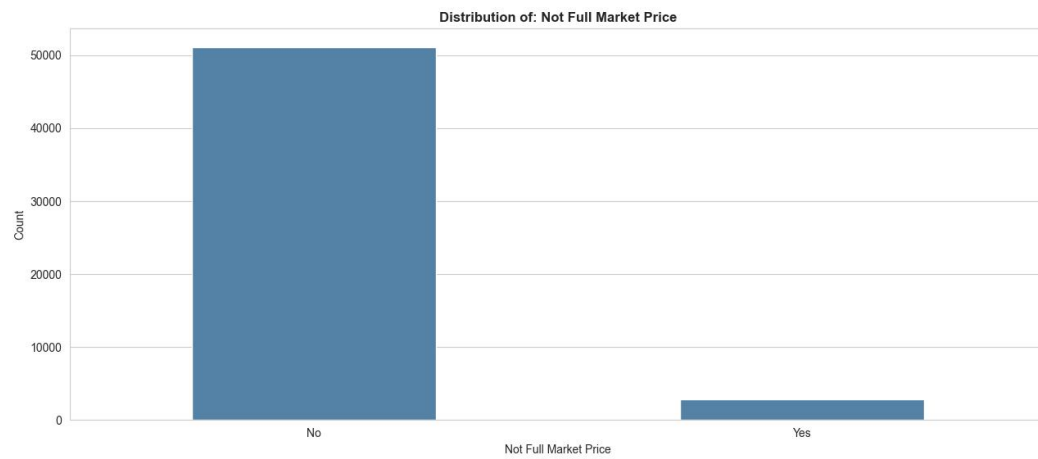
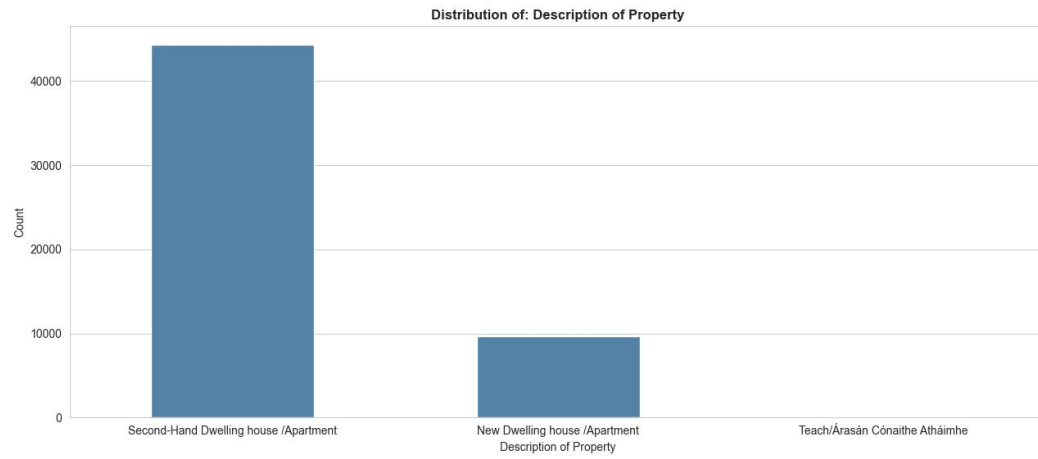
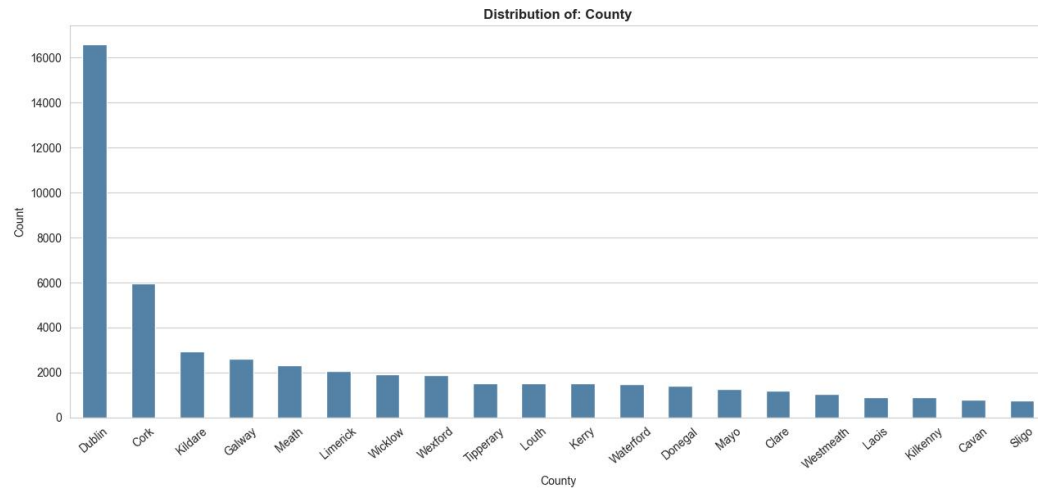
1. Section mapping: 1.7 Missing Value Visualisation
2. Basic description: Plot generated during DQR exploratory checks to support data quality interpretation.



Figure A2: Output from cell 31 (1.8 Categorical Feature Distributions)

3. Section mapping: 1.8 Categorical Feature Distributions
4. Basic description: Plot generated during DQR exploratory checks to support data quality interpretation.

Categorical Feature Distributions



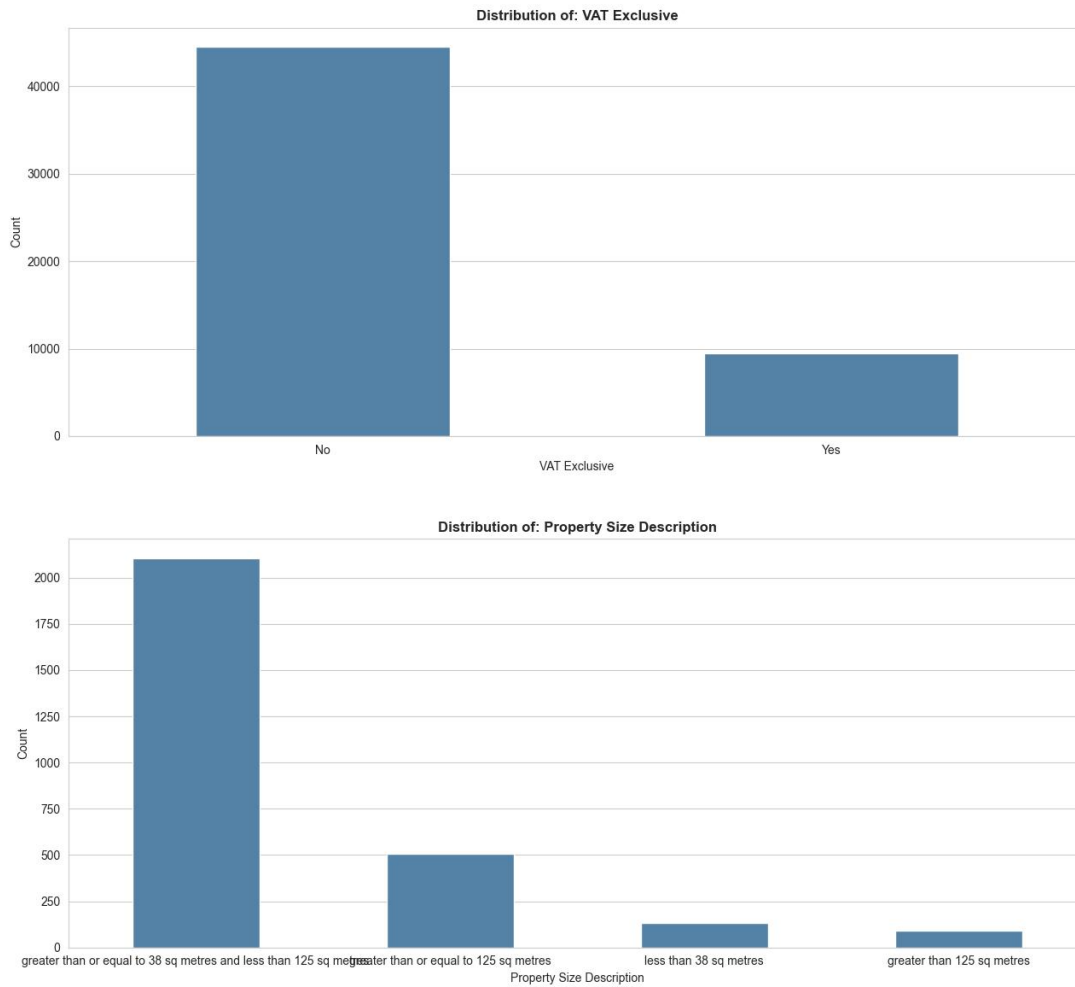


Figure A3: Output from cell 35 (1.9 Price Distribution Analysis)

5. Section mapping: 1.9 Price Distribution Analysis
6. Basic description: Plot generated during DQR exploratory checks to support data quality interpretation.

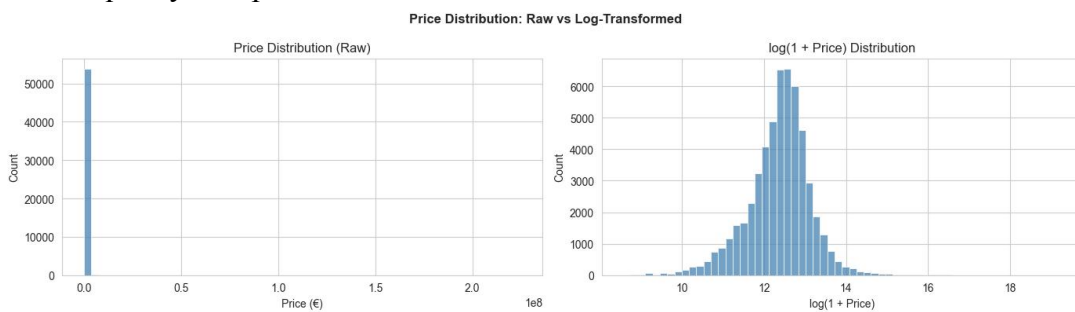
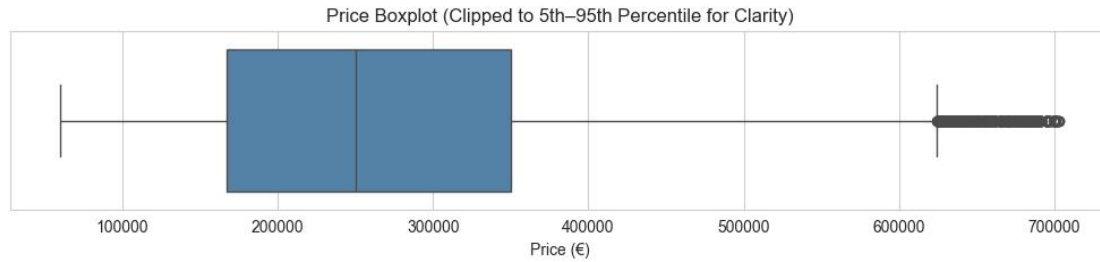


Figure A4: Output from cell 35 (1.9 Price Distribution Analysis)

7. Section mapping: 1.9 Price Distribution Analysis
8. Basic description: Plot generated during DQR exploratory checks to support data quality interpretation.



A.2 Tables (Analysis Outputs Only)

Table A1: Initial Data Structure Check (cell 6)

- 9. Section mapping: 1.2 Initial Data Structure Check
- 10. Basic description: Missingness and structure diagnostics (e.g., `display(missing_df)`).

Table A2: Initial Data Structure Check (cell 6)

- 11. Section mapping: 1.2 Initial Data Structure Check
- 12. Basic description: Complementary structure summary table from the same step.

Table A3: Data Type Conversion Summary (cell 9)

- 13. Section mapping: 1.3 Data Type Conversion
- 14. Basic description: Field-level raw-to-converted dtype summary.

Table A4: Numeric Features Summary (cell 19)

- 15. Section mapping: 1.6.1 Numeric Features Summary Table
- 16. Basic description: Descriptive statistics for numeric features.

Table A5: Categorical Features Summary (cell 22)

- 17. Section mapping: 1.6.2 Categorical Features Summary Table
- 18. Basic description: Cardinality/top-frequency summary for categorical features.

Table A6: High-Cardinality Feature Summary (cell 25)

- 19. Section mapping: 1.6.3 High-Cardinality String Features
- 20. Basic description: High-cardinality diagnostics for string features.

Table A7: Price Outlier Investigation Table (cell 38)

- 21. Section mapping: 1.10 Price Outlier Investigation
- 22. Basic description: Percentile/boundary-based outlier summary.

Table A8: VAT Consistency Check Table (cell 42)

- 23. Section mapping: 1.13 VAT Consistency Check
- 24. Basic description: VAT vs property-type consistency cross-check.

Table A9: VAT Consistency Check Supplement (cell 42)

- 25. Section mapping: 1.13 VAT Consistency Check
- 26. Basic description: Additional table output from VAT rule diagnostics.

A.3 DQR CSV Exports from project DQR section

27. `organize/train-cleaned.csv`
28. `organize/DQR-NumericFeatures.csv`
29. `organize/DQR-CategoricalFeatures.csv`